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EXP.NO:1 Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

CODE:

```
data = [
    ["Sunny", "Warm", "Normal", "Strong", "Warm", "Same", "Yes"],
    ["Sunny", "Warm", "High", "Strong", "Warm", "Same", "Yes"],
    ["Rainy", "Cold", "High", "Strong", "Warm", "Change", "No"],
    ["Sunny", "Warm", "High", "Strong", "Cool", "Change", "Yes"]
]
hypothesis = ["0"] * (len(data[0]) - 1)
print("Initial Hypothesis:")
print(hypothesis)
for example in data:
    if example[-1] == "Yes":
        for i in range(len(hypothesis)):
            if hypothesis[i] == "0":
                hypothesis[i] = example[i]
            elif hypothesis[i] != example[i]:
                hypothesis[i] = "?"
print("\nFinal Hypothesis:")
print(hypothesis)
```

OUTPUT:

```
Initial Hypothesis:
['0', '0', '0', '0', '0', '0']

Final Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']
```